Universidad Autónoma de Madrid

Cookies and their Effect on Privacy in Web Browsing

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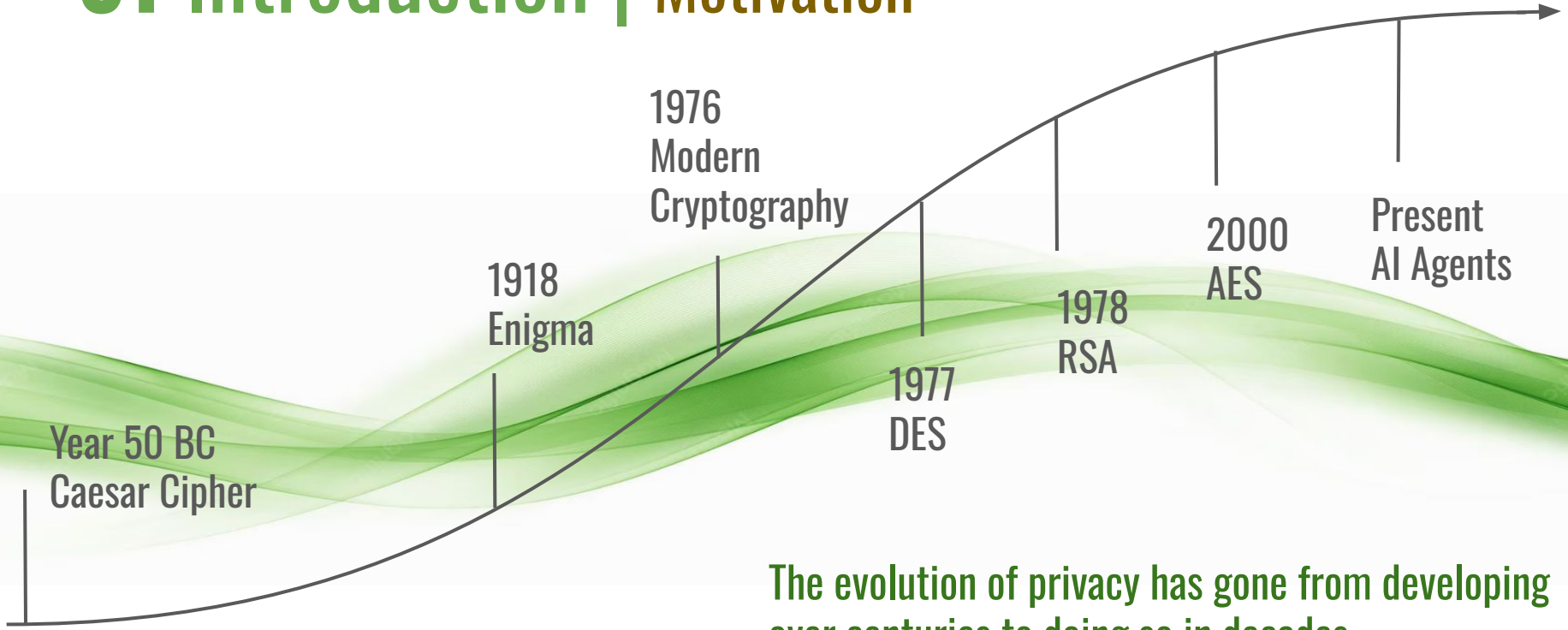
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01
Introduction

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01 Introduction | Motivation



The evolution of privacy has gone from developing over centuries to doing so in decades

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01 Introduction | Objectives



1. Design and implement a controlled environment to automatically capture web traffic under different cookie consent scenarios.
2. Analyze the generated traffic to identify patterns related to DNS queries, TLS connections, and communications with external services.
3. Study the distribution of domains present in the traffic, distinguishing between highly frequent domains and less frequent domains.
4. Evaluate the impact of cookie acceptance or rejection on observed traffic patterns.
5. Compare different feature representation techniques to analyze their utility in the study of encrypted traffic.
6. Train and evaluate machine learning models capable of identifying visited domains from captured traffic.
7. Determine whether encrypted traffic metadata contains sufficient information to infer a user's browsing activity, even when cookies are rejected.

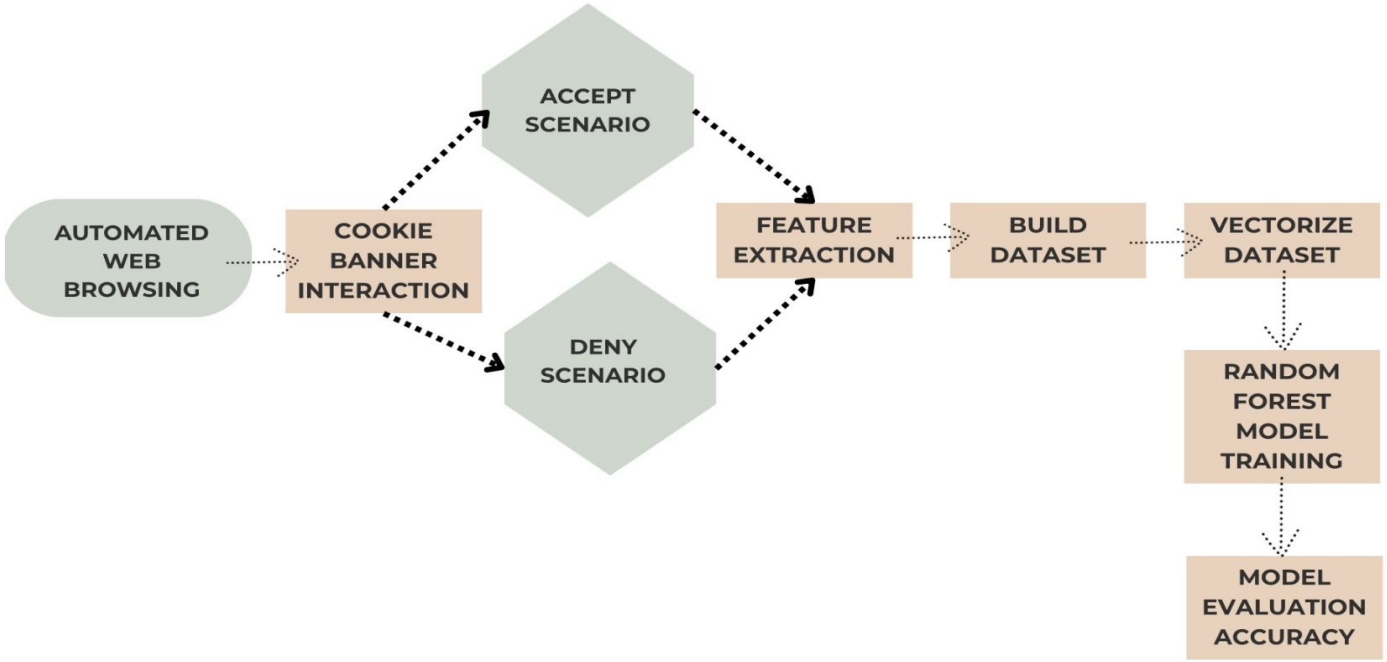
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02
Methodology

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02 Methodology

The following figure illustrates the methodology employed for traffic capture, feature extraction, and machine learning model evaluation



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02 Methodology

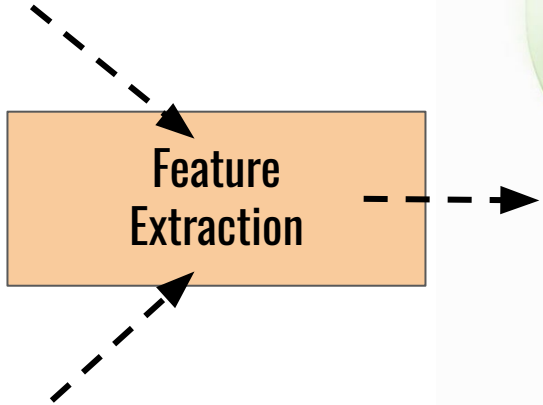
A browsing session associated with YouTube could generate the following sequences:

```
Domain:
youtube.com

Server IP sequence:
142.250.184.78 142.250.184.110 172.217.168.14 ...

DNS query sequence:
youtube.com googlevideo.com ytimg.com
youtubei.googleapis.com doubleclick.net ...

SNI sequence:
youtube.com googlevideo.com ytimg.com ...
```



Since machine learning models cannot directly process textual data, the next step involves converting these sequences into numerical representations using the HashingVectorizer.

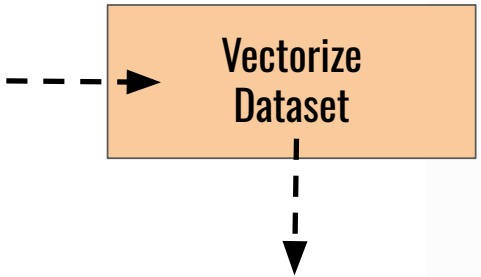
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02 Methodology

The HashingVectorizer transforms each token into a fixed numerical index using a hash function. Conceptually, the preceding sequences could be transformed into something similar to the following:

```
Hashed token representation:

youtube.com                 → 12
googlevideo.com             → 1239
ytimg.com                   → 2131
youtubei.googleapis.com     → 323
doubleclick.net             → 10
```



After applying the hash function to all tokens, the vectorizer generates a sparse numerical vector representing the frequency of the evaluated (hashed) features within the browsing session.

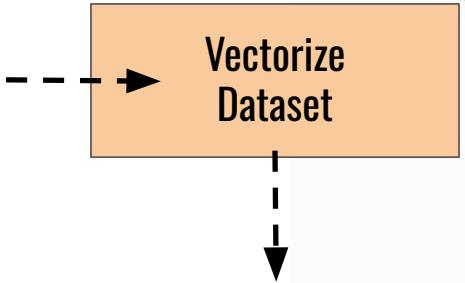
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02 Methodology

This vector contains the frequency distribution of hashed tokens within a high-dimensional feature space.

```
Sparse hashed vector:

[0, 1, 0, 2, 0, 0, 1, 0, 0, 3, ...]
```



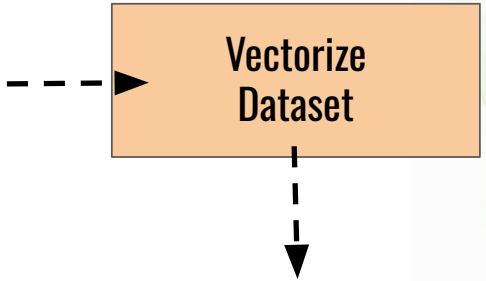
Before adopting the hashing-based representation, embedding-based approaches were also experimentally evaluated. In that case, each domain token was converted into a dense numerical vector designed to capture semantic similarities between the elements.

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02 Methodology

A simplified example of what the machine learning model receives would be:

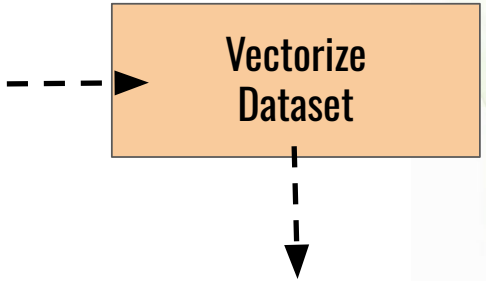
```
[
  total_packets = 542,
  total_bytes = 182394,
  avg_packet_size = 336.5,
  tcp_packets = 498,
  dns_packets = 12,
  duration_seconds = 18.2,
  ip_12 = 1,
  ip_323 = 2,
  dns_1239 = 3,
  sni_2131 = 1,
  ...
]
```



This final numerical representation encodes both the statistical properties of the traffic and the structural communication patterns associated with the browsing session. Subsequently, the Random Forest model uses these vectors to learn discriminative patterns capable of identifying the visited domain.

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02 Methodology



Another simplified alternative that was considered would be:

Embedding representation:

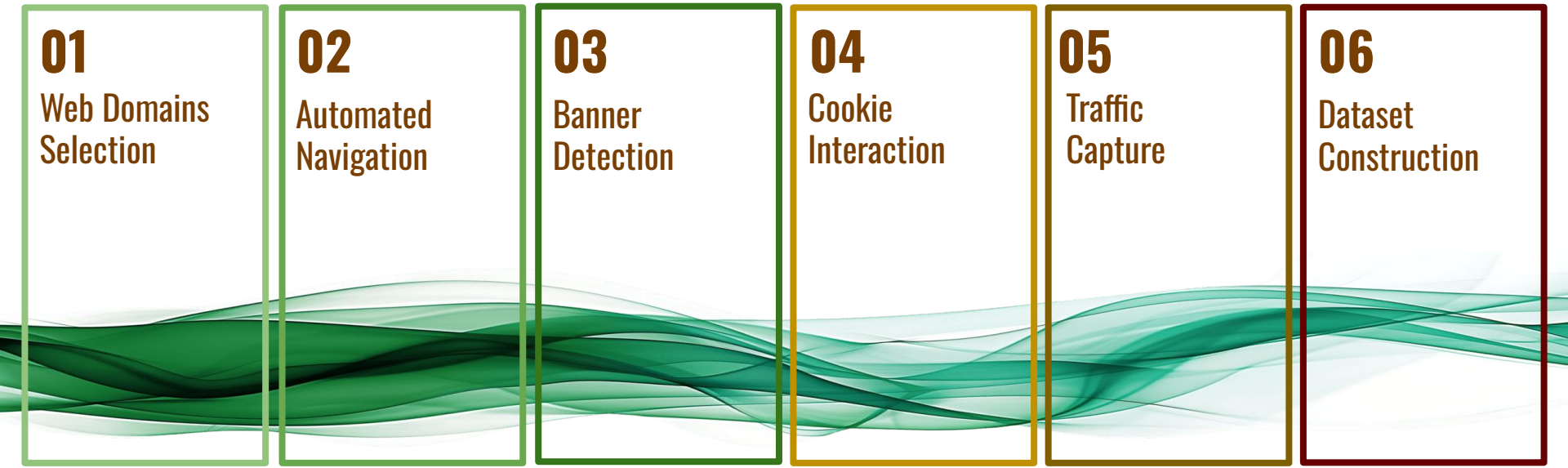
```
youtube.com          → [9.1, -1.2, 0.4, ...]
googlevideo.com      → [8.1,  2.0, -3.0, ...]
ytimg.com            → [4.2, -0.7, 1.8, ...]
```

Finally, all the extracted information is merged into a single feature vector, along with the numerical traffic statistics extracted from the PCAP capture. The resulting representation constitutes the final input fed into the Random Forest classifier.

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02 Methodology | Web Browsing Data

The cookie consent interaction methodology was implemented in this way.



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02 Methodology | Web Browsing Data

In this way, the processing of the packets on the network was carried out.



Automated browsing sessions were conducted on the selected websites, interacting with cookie banners under both acceptance and rejection scenarios, while all network traffic was captured in separate PCAP files.



Each PCAP file was processed to generate a structured representation of the session, extracting metrics such as traffic volume, number of packets, protocol activity, and communication duration.



In addition to general statistics, TCP flows were analyzed, and ordered sequences of IP addresses, DNS queries, and SNI values were extracted to capture browsing behavior.



Finally, all numerical and sequential features were combined into a single record per session, generating datasets prepared for the training and evaluation of machine learning models.

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02 Methodology | Data Analysis and Feature Extraction

Why was the vectorization performed?

Sequences of IP addresses, DNS queries, and SNI values could not be used directly in the models, as they were represented as text and had variable lengths.

How was it resolved?

To solve this problem, vectorization was applied, transforming each sequence into a fixed-size numerical vector regardless of the number of observed elements.

Embedding dimension

A dimensionality of 2^{12} (4096) features was used, seeking a balance between representation capacity, computational cost, and the probability of collisions between distinct elements.

Benefits

This strategy enables scaling to large volumes of domains and network identifiers without the need to maintain an explicit vocabulary, adapting to the constant evolution of web traffic.

Final set

Finally, the obtained vectors were combined with the statistical characteristics of the traffic to construct the final dataset used by the machine learning models.

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02 Methodology | Data Analysis and Feature Extraction

1

Random Forest was selected for its robustness and ability to handle high-dimensional data combining statistical and sequential features.

Each browsing session is represented by traffic statistics and embeddings derived from IP, DNS, and SNI sequences.

2

3

The data were split into training and testing sets using an 80/20 ratio, keeping captures from the same execution separate.

The 20 most frequent domains were retained as individual classes, and the rest were grouped into a class named 'other'.

4

5

Performance was evaluated using accuracy, comparing cookie acceptance and rejection scenarios.

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03
Results

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03 Results | Elephants and Mice

1. Traffic Distribution and Domain Persistence

Elephants

Domains that generate large volumes of traffic and appear persistently across multiple browsing sessions, generally associated with shared web infrastructures and services.



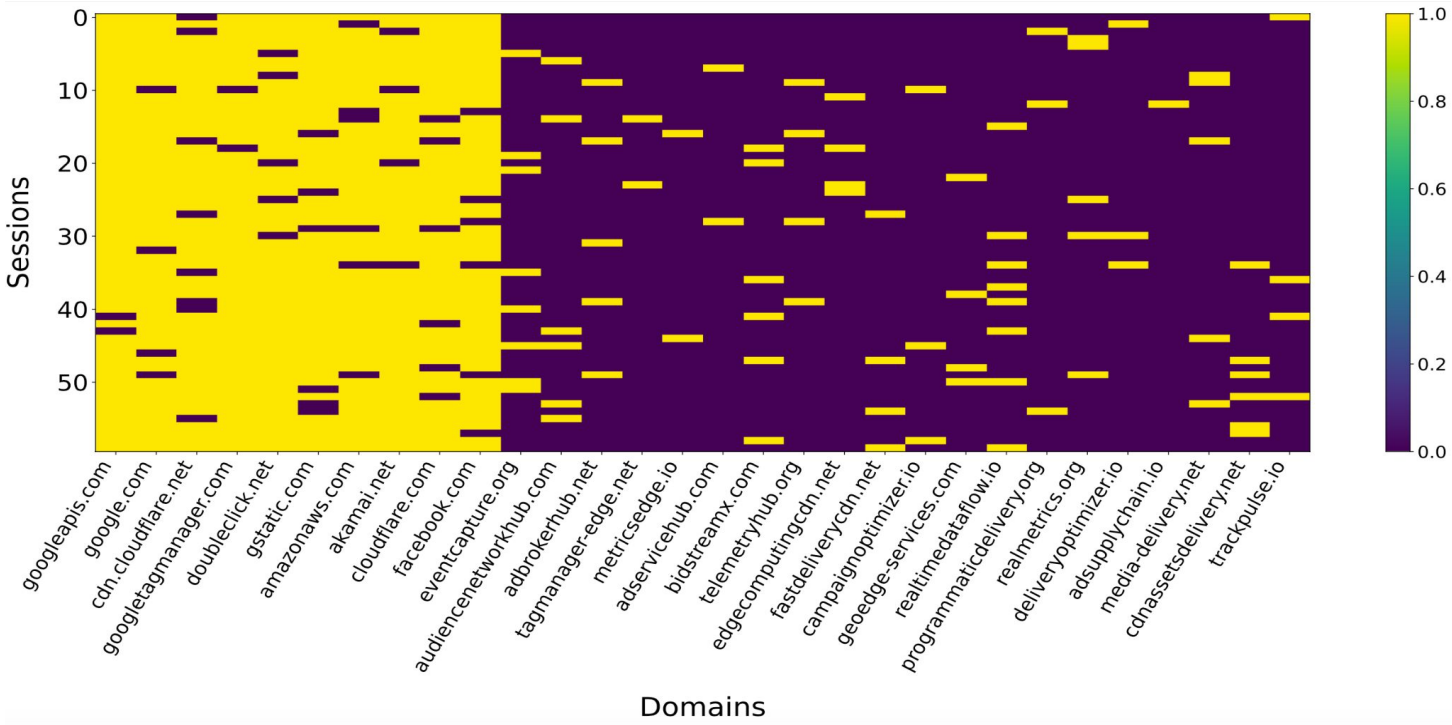
Domains that appear sporadically and depend on the website visited, providing specific information about the user's browsing activity.

Mice

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03 Results | Elephants and Mice

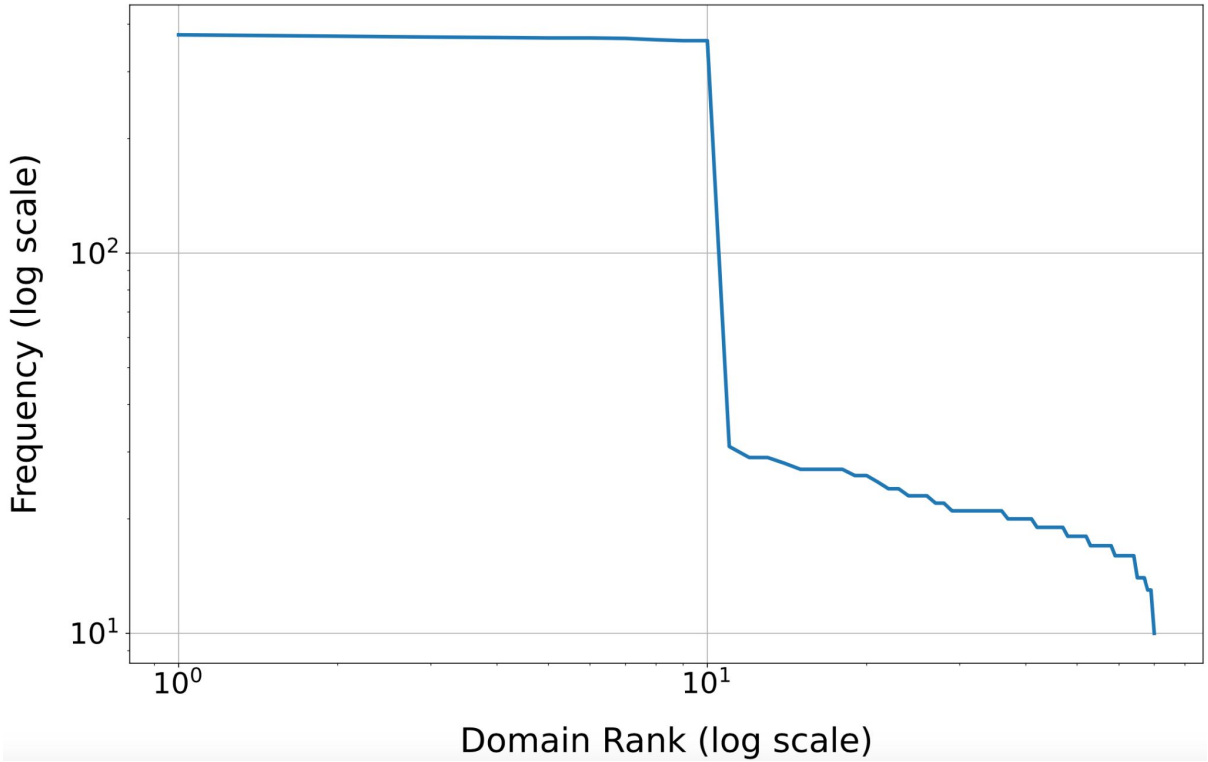
El siguiente mapa de calor muestra la presencia de los distintos dominios a lo largo de las sesiones de navegación capturadas.



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03 Results | Elephants and Mice

La siguiente gráfica representa la evidencia cuantitativa de este fenómeno



The majority of traffic is concentrated in a few persistent domains, while the rest appear sporadically.

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03 Results | Elephants and Mice

2. The Vocabulary Problem

01 The DNS, IP, and SNI sequences contain very frequent domains (Elephants) and thousands of infrequent domains (Mice).

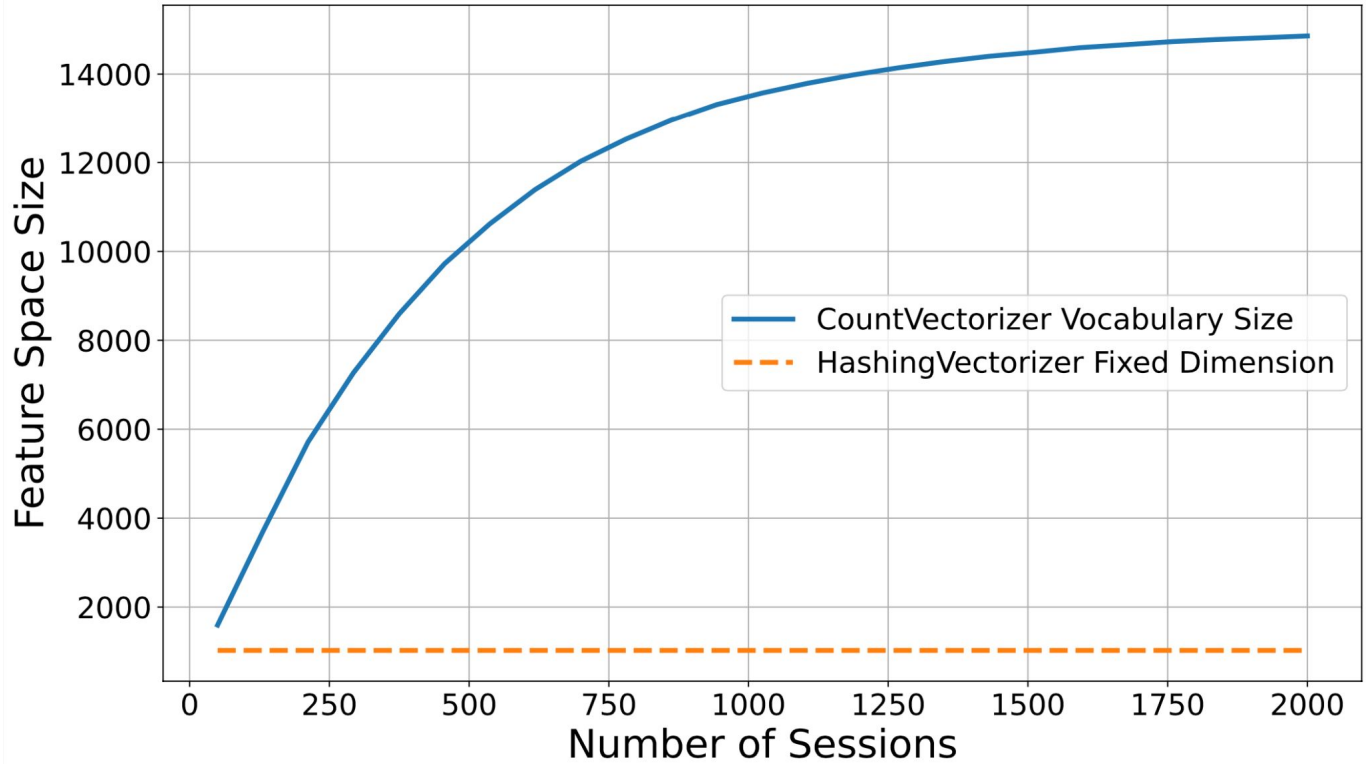
02 Selecting the embedding technique is critical, as the representations must be robust against repetitive patterns and highly imbalanced distributions.

03 HashingVectorizer maintains a fixed dimensionality, preventing uncontrolled vocabulary growth and improving scalability.

04 Other vectorization options, such as CountVectorizer and SentenceTransformers, have been considered.

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03 Results | Elephants and Mice



The number of observed domains grows continuously with browsing sessions. While CountVectorizer increases its dimensionality to incorporate new domains, HashingVectorizer maintains a fixed and scalable feature space.

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03 Results | Accept vs. Deny vs. No Action

Objective

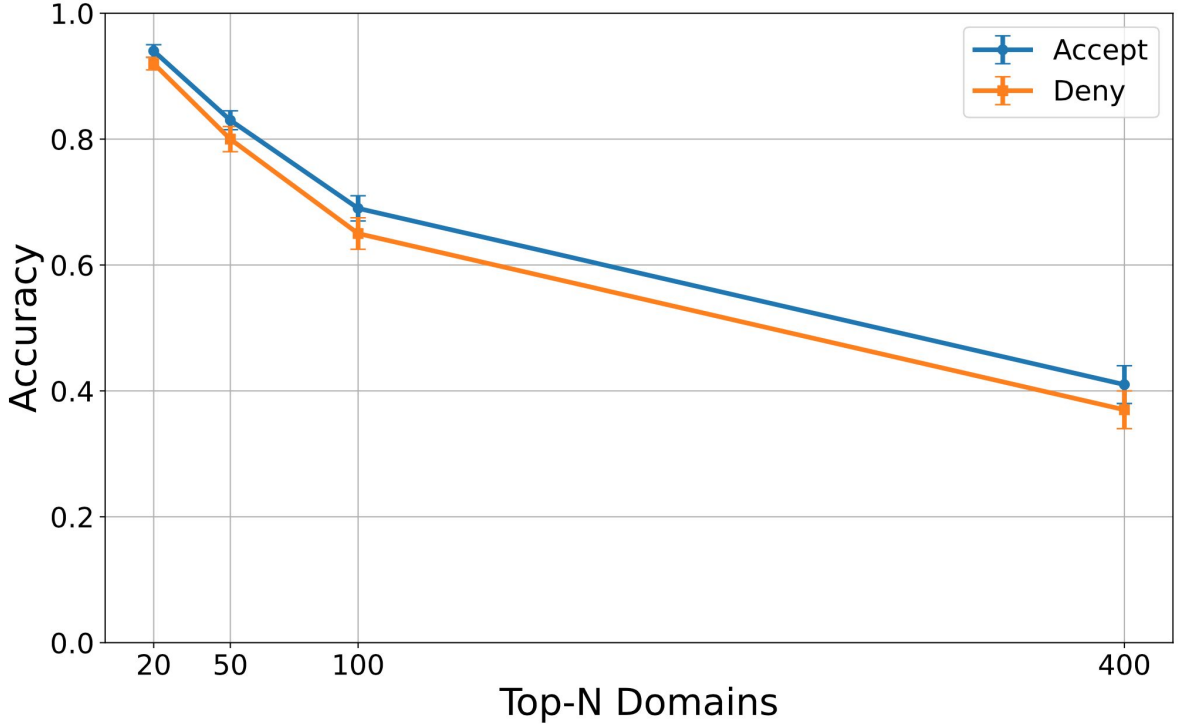
Assess whether the decision to accept or reject cookies affects the ability to identify the website visited.

Main Results

- No significant differences were observed between the Accept and Deny scenarios.
- Accuracy decreases as the number of domains (Top-N) increases, due to the greater complexity of the problem.
- DNS, SNI, IPs, and communication patterns continue to provide sufficient information to identify the visited domain.
- Some sessions correspond to a "No Action" scenario due to banners being absent or not automatically detected.

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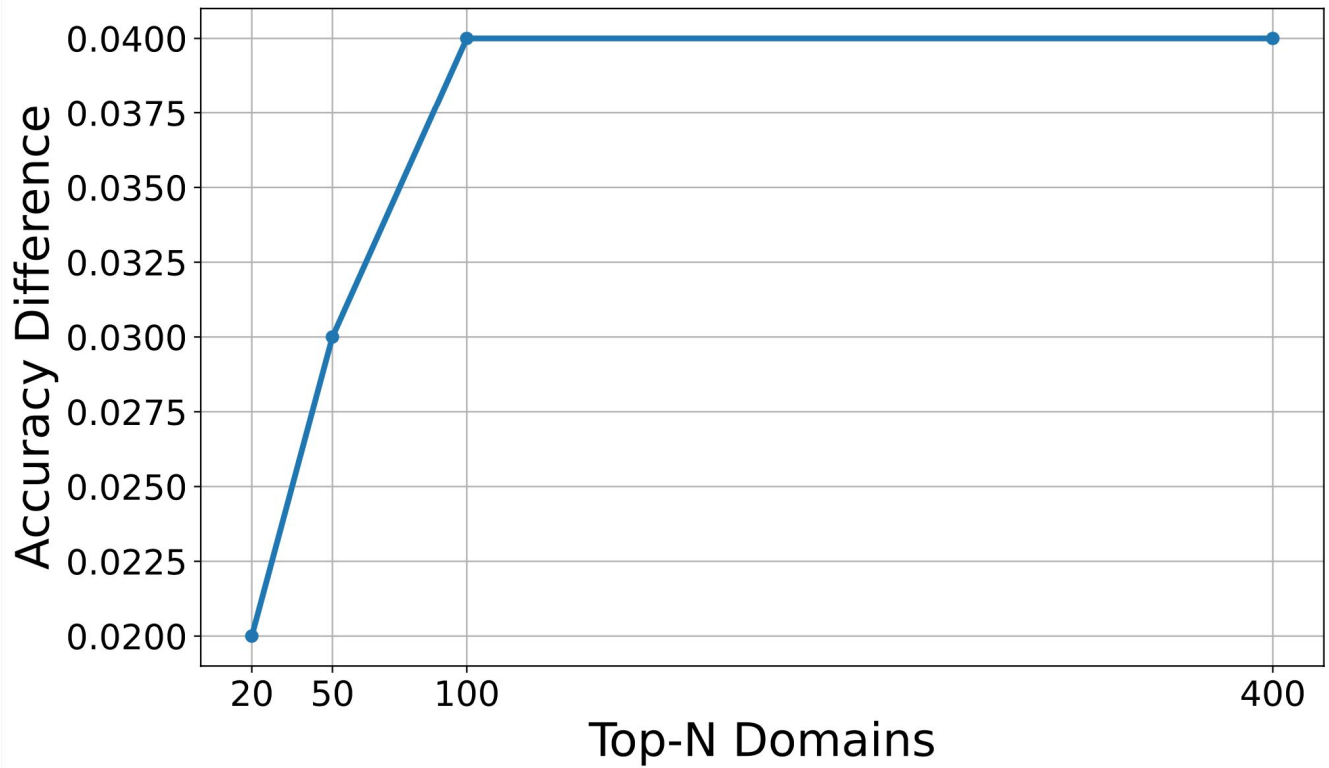
03 Results | Accept vs. Deny vs. No Action



Precision decreases as the number of domains considered increases, reflecting the growing complexity of the classification problem. However, the results obtained for Accept and Deny remain very close across all the configurations analyzed.

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03 Resultados | Accept vs. Deny vs. No Action



The difference in accuracy between the two scenarios remains below 4% even in the most complex configurations, indicating that cookie consent has a limited impact on the ability to identify the visited domain.

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03 Results | Impact of Classification Complexity

Impact of complexity

The complexity of the problem increases as the number of domains (Top-N) considered in the classification rises.

Evolution of precision

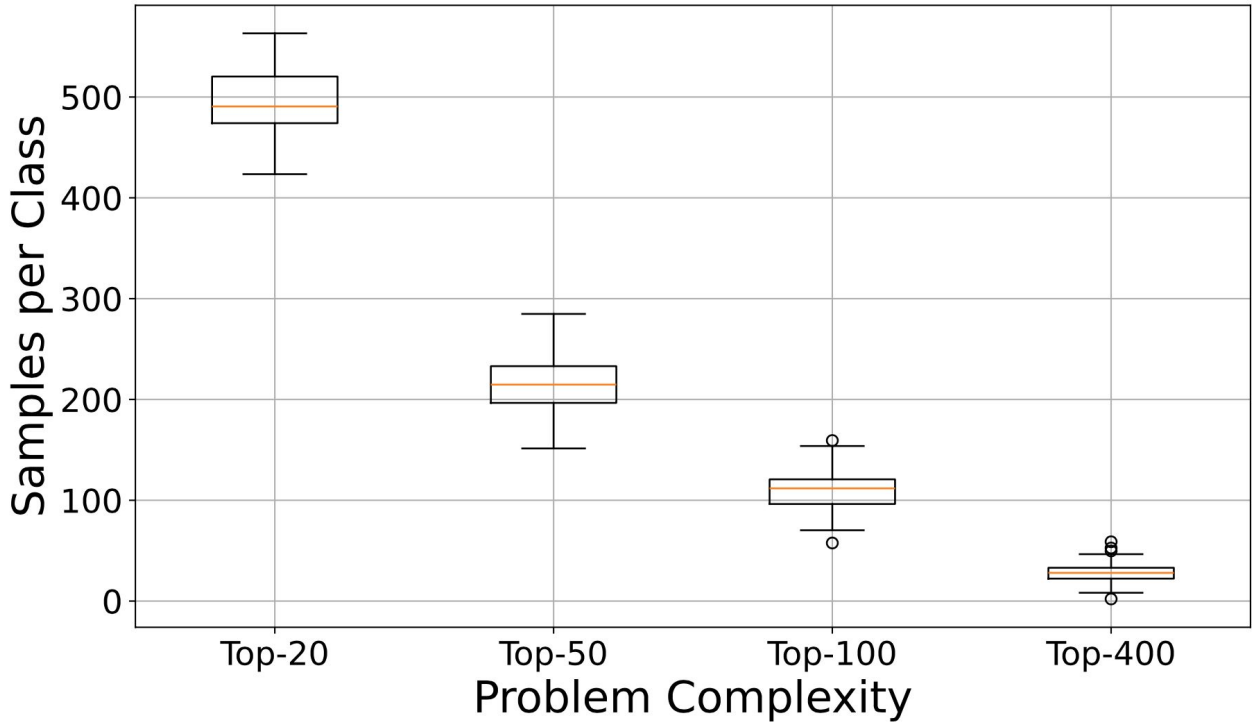
Precision decreases progressively when moving from the Top 20 to the Top 400, reflecting the greater difficulty in distinguishing between an increasing number of websites.

Reason/Motivation

As the number of classes increases, the available samples per domain decrease and similarities between traffic patterns increase, making it difficult to identify clear decision boundaries.

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03 Results | Impact of Classification Complexity



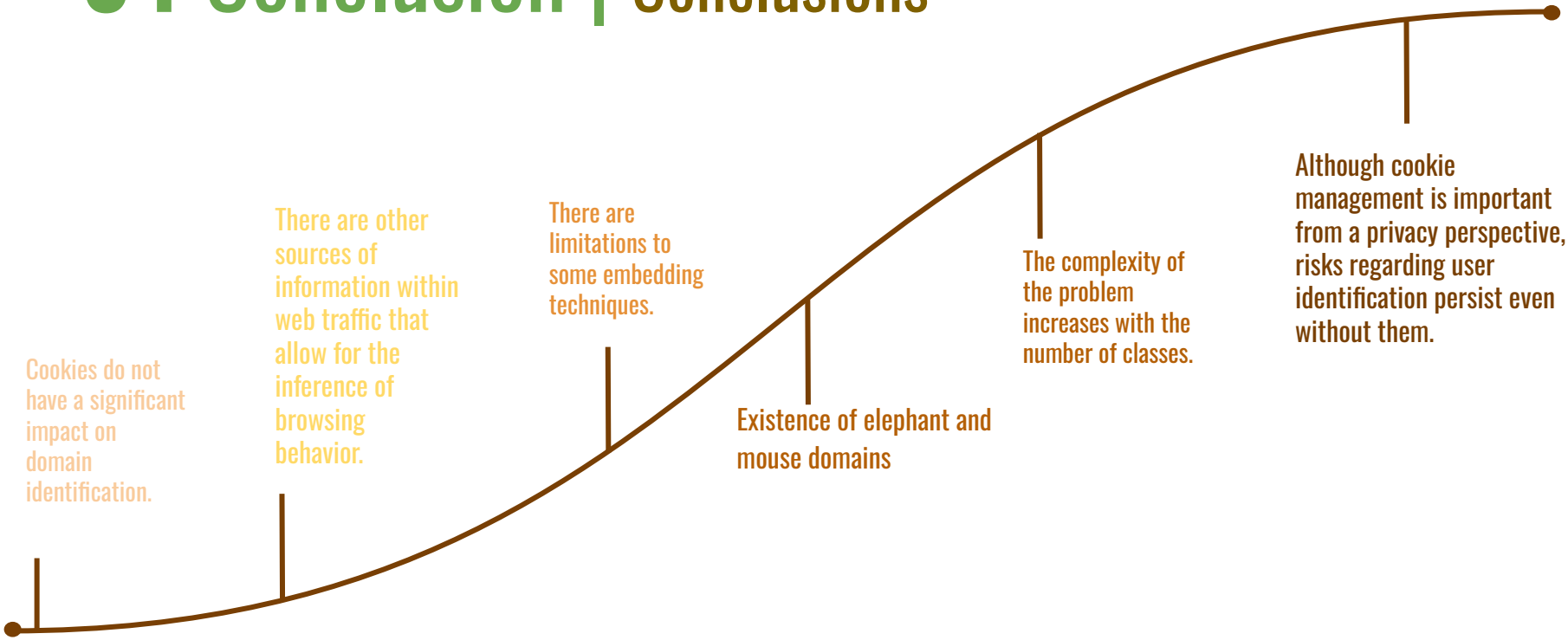
The model's accuracy progressively decreases as the number of domains considered increases. As the complexity of the problem grows, traffic patterns between different websites become more similar, making classification more difficult.

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04
Conclusion

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04 Conclusion | Conclusions



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04 Conclusion | Future Work

Advanced cookie banner automation

Use of AI agents and browsers with MCP to improve the detection of and interaction with consent banners.

New embedding techniques for network traffic

Design representations that distinguish between elephant domains (common and frequent) and mice domains (specific to each website).

Temporal representations of traffic

Incorporate models that take into account the order and timing of requests made during navigation.

More advanced learning models

Evaluate techniques such as gradient boosting, neural networks, and sequence- or graph-based models.

More realistic navigation scenarios

Combine automated sessions with real user browsing and consider different devices, browsers, and geographic locations.

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Questions

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